Latitudinal coherence of Meridional Heat Transport in the Atlantic

Drivers of variability

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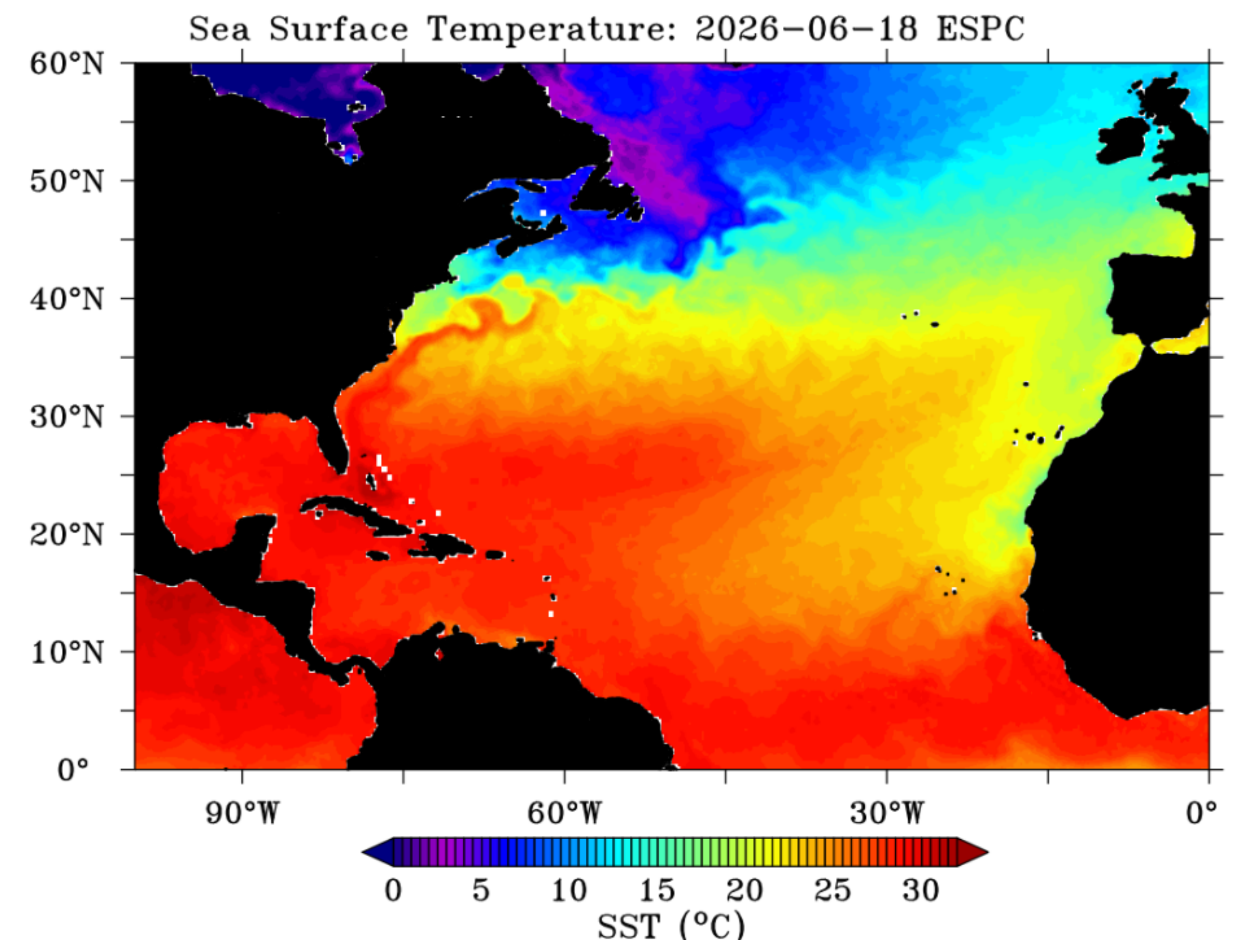
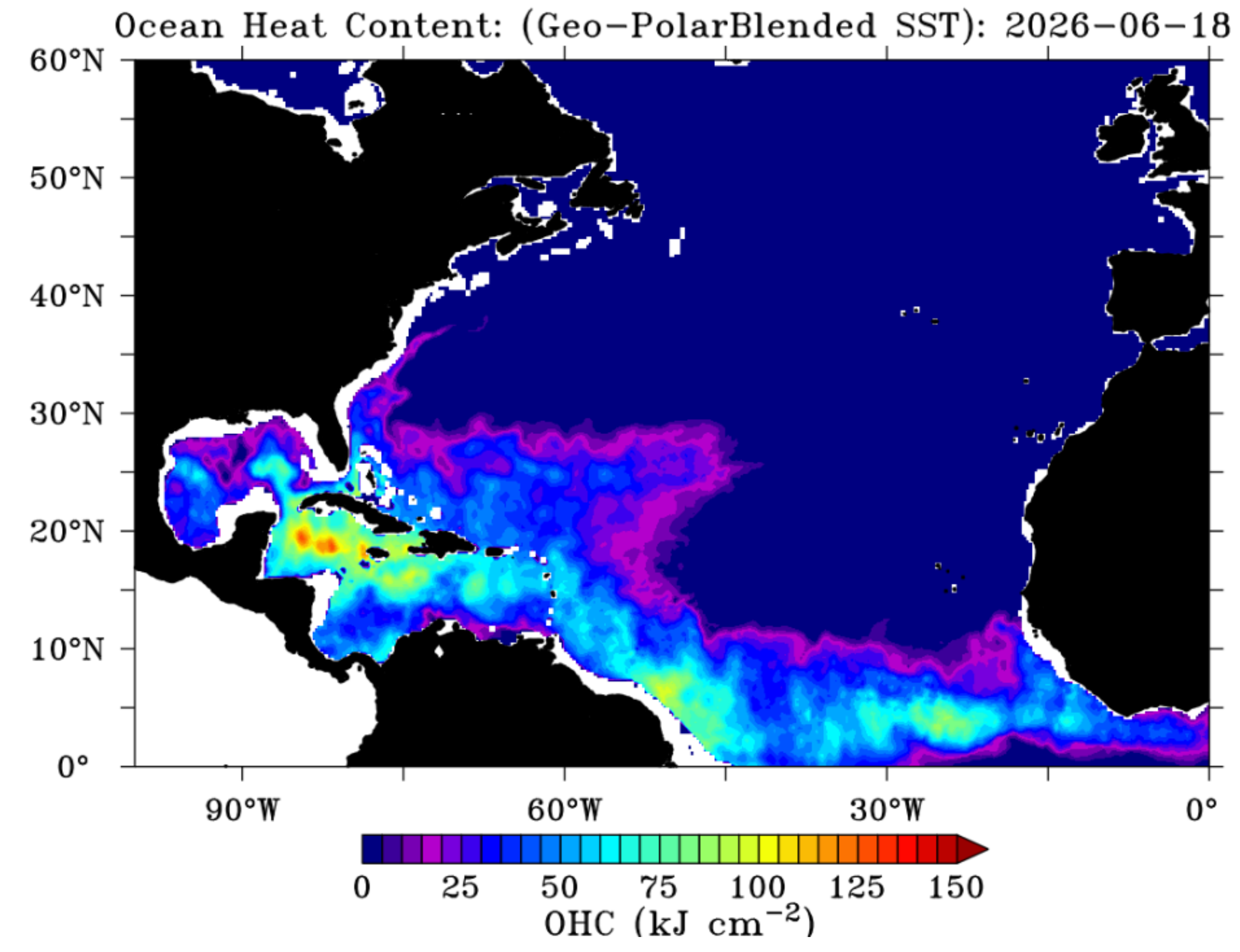
Outline - Content

1. What is heat transport and why do we care about it?
2. What is the research gap?

   MD klein (für Konsistenz)
3. Thesis Objectives
4. Data & Methods
5. First look at the data
6. Next steps





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Motivation

klein ⇒ nach „:" groß

groß

Earth's Energy Budget in Imbalance: where does the excess energy go?

↳ klein

- Imbalance: currently more energy absorbed than emitted

→ from where

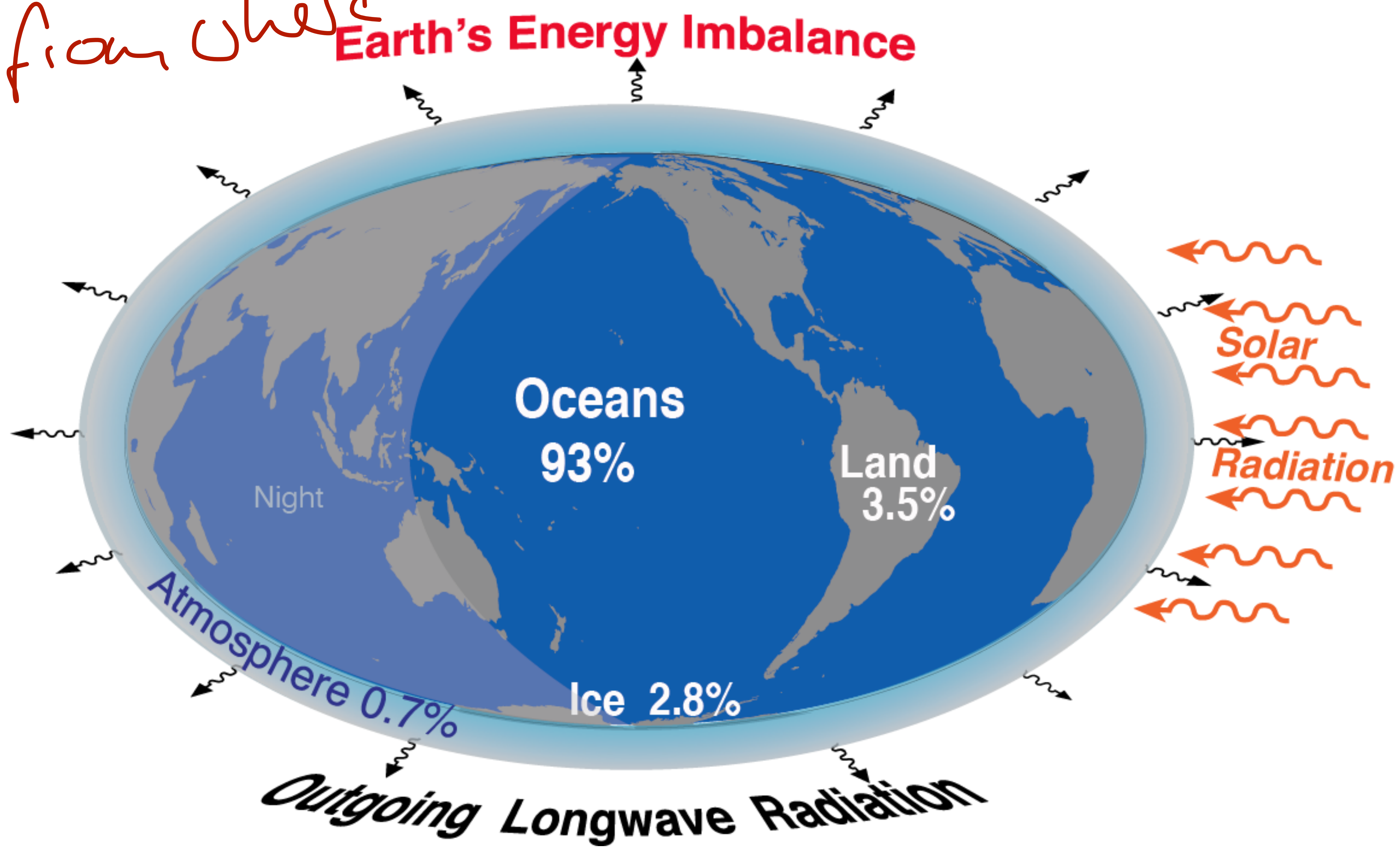
→ warum currently

- Ocean is dominant heat sink with an uptake of 93% of this imbalance energy

→ This excess heat enters the ocean via surface heat flux (F_S)

But what happens to this heat once its in the ocean?

it's



Wikipedia

Motivation

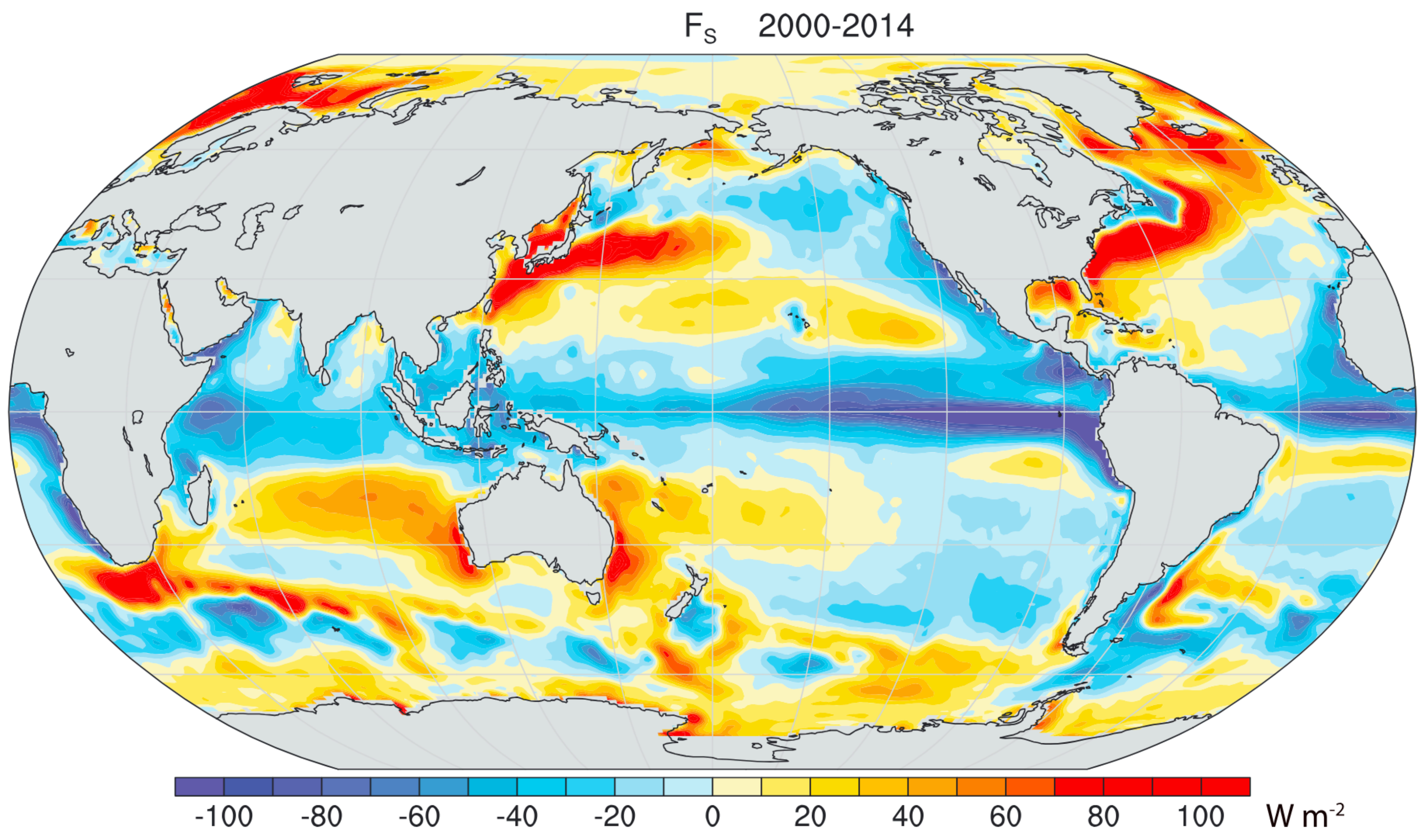
From surface heat flux to ocean heat transport

→ Formel tüchm?

- Surface heat flux F_S - energy (heat) is absorbed in the ocean
- This heat is either stored or recirculated

(*F_S* here upwards)

$$F_s = -\mathrm{dOHC}/dt - \nabla\cdot\mathbf{F_O}$$



Surface heat flux (net upward) for 2000-2014 in W/m^2

dOHC/dt: change in ocean heat content, >0 ⇒ ocean warming

∇F_O: heat transport convergence (HTC), >0 ⇒ divergence, transport away

⇒ Rearranging for ∇F_O & zonally integrating gives an equation for **MHT** from the energy budget:

$$\mathrm{MHT}(\phi) = \int_{\phi}^{90} [F_s + \mathrm{dOHC}/\mathrm{dt}]\, a d\phi$$

Equations and Image from *Trenberth and Fasullo (2017)*

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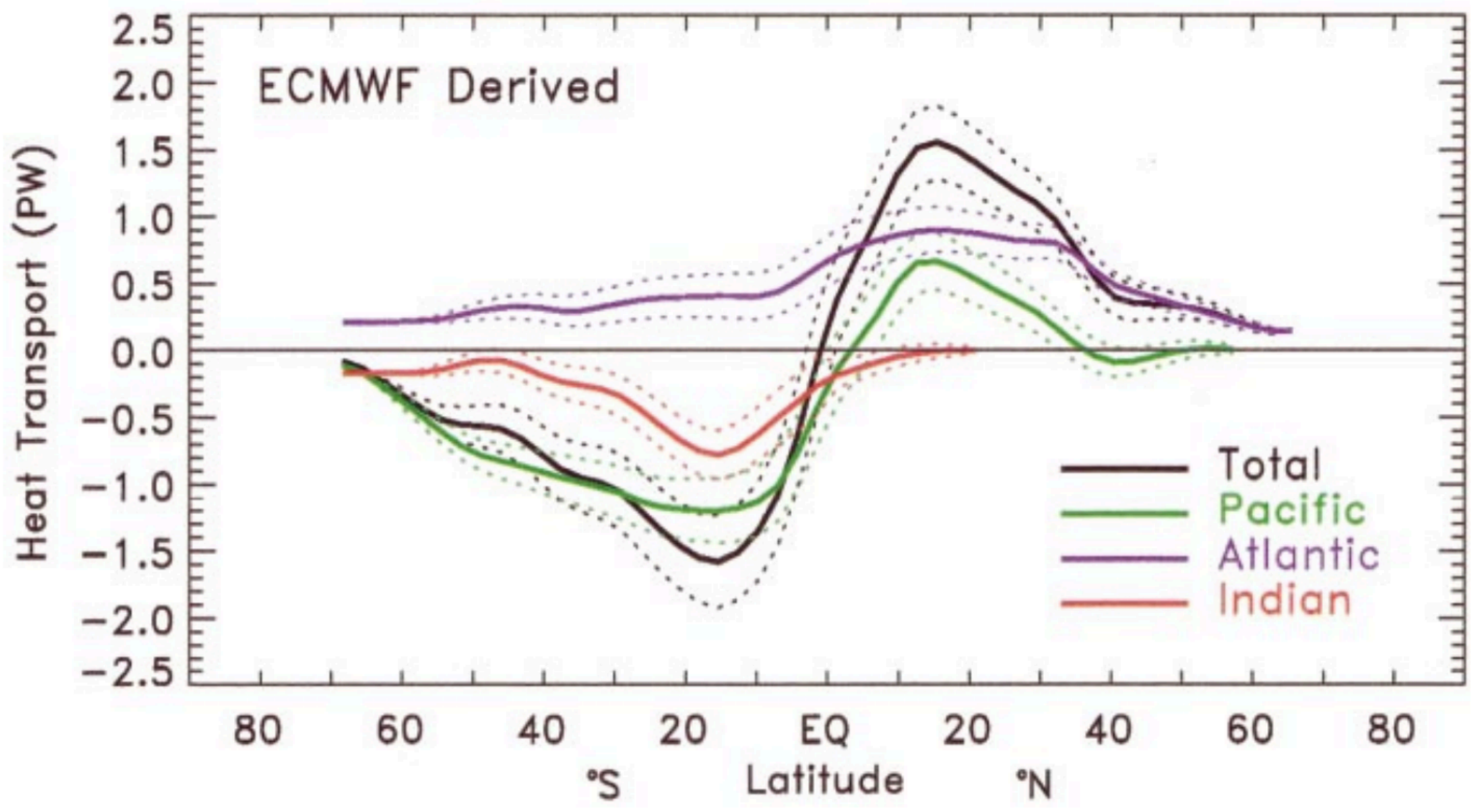
Motivation

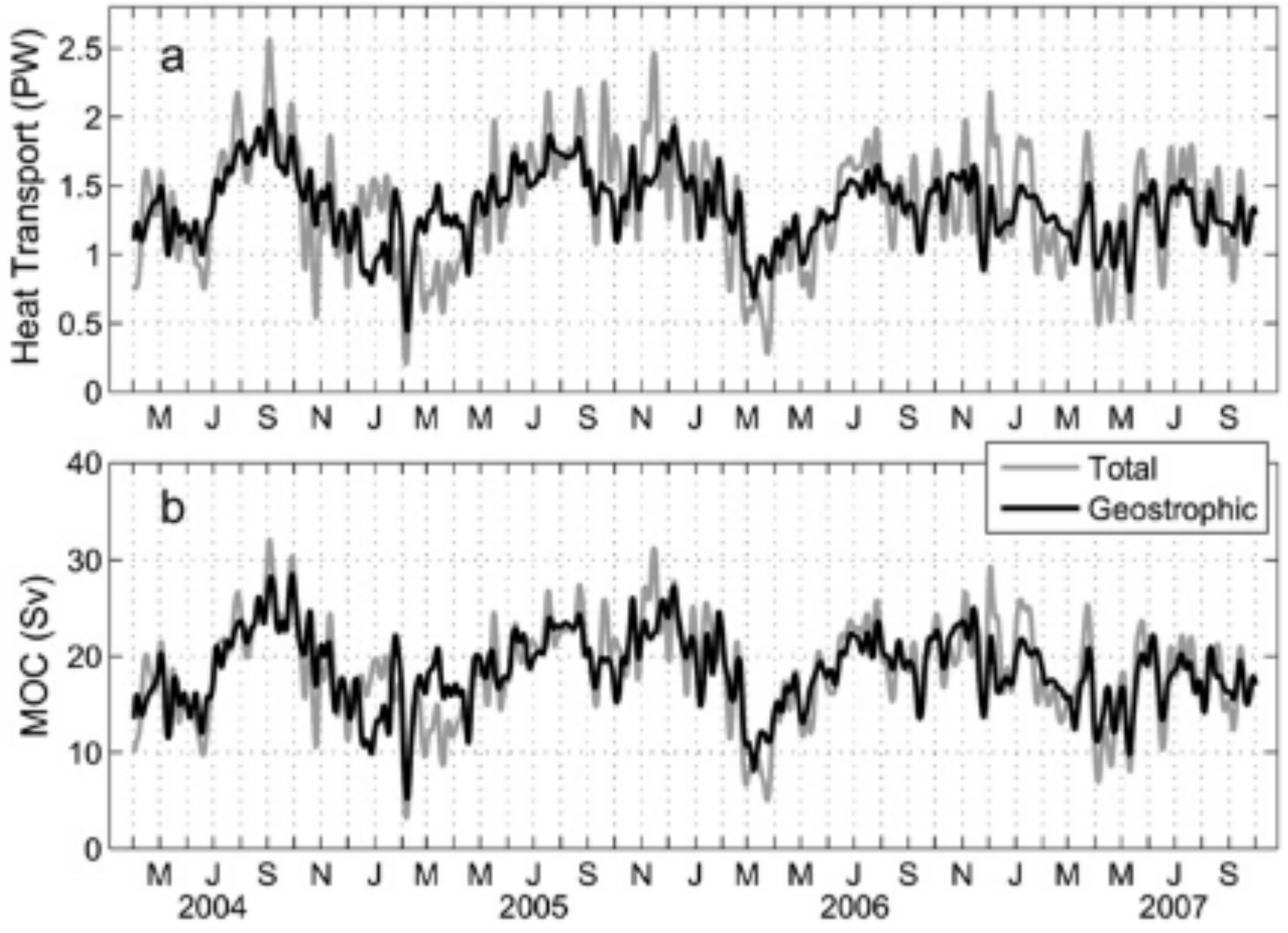
aber ich mag die Subtitle

The Atlantic: ~~a~~ A special case

- This happens globally but:
- **Atlantic is unique:** ~~h~~Heat transport is northward at all latitudes
- This transport is dominated by the **AMOC** (Atlantic Meridional Overturning Circulation)



Global estimates for heat transport, *Trenberth and Caron (2001)*



Heat Transport and MOC at 26°N, *Johns et al. (2011)*

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Motivation

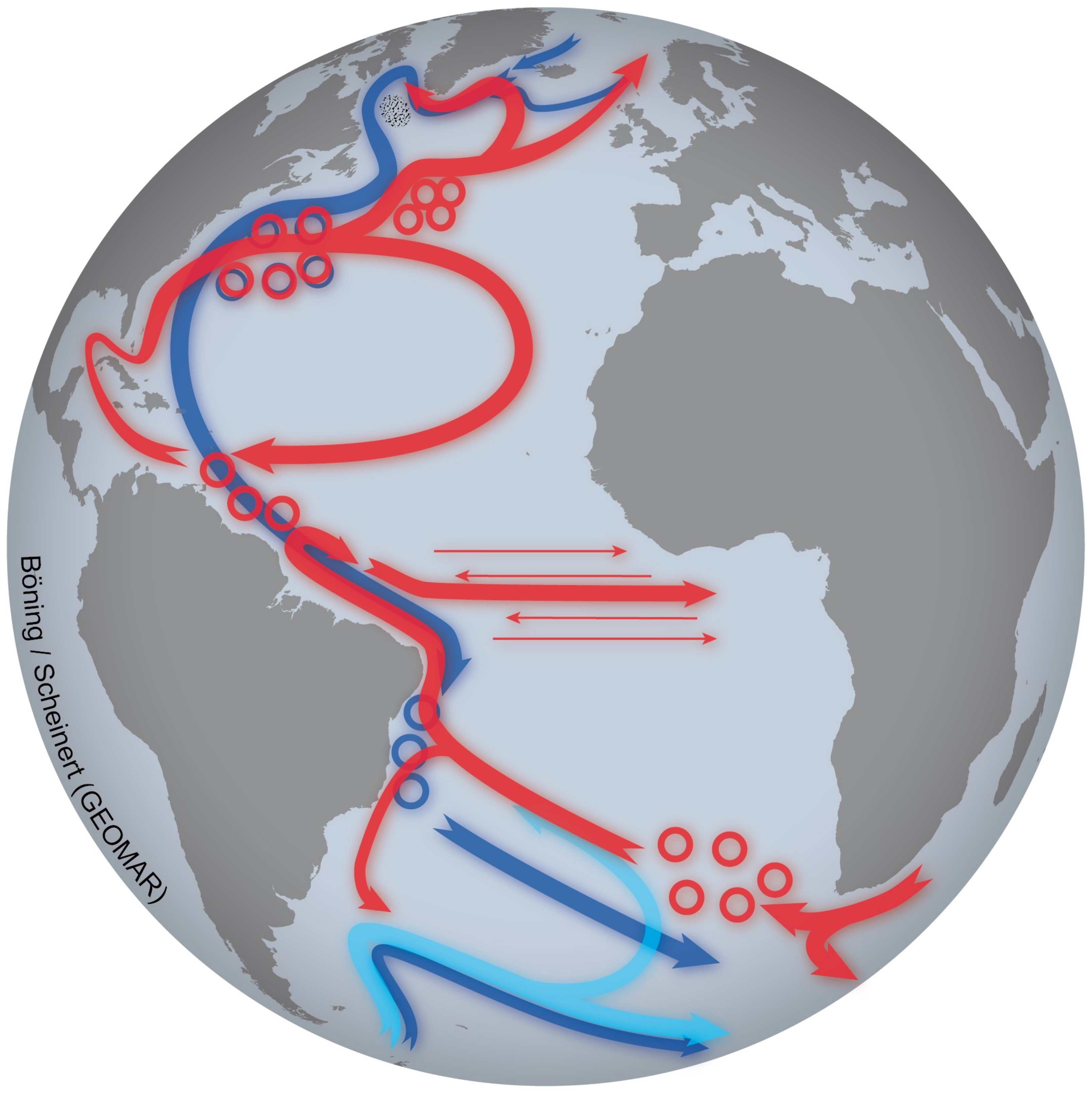
What is the AMOC?

AMOC:

- Warm surface ~~current~~ water flows/transported northward
- Deep convection close to arctic where surface water cools and sinks
- Transported back south at depth

- Connects with gyre circulations and eddies etc.
- ~~Covers the whole Atlantic ocean basin~~
- ~~r~~Responsible for most heat (and carbon) transport in NH (Europe has milder climate than other location on same longitudes)

=> Vielleicht als neuer eingerückter Punkt?



www.geomar.de

Motivation

The Atlantic: a special case

A

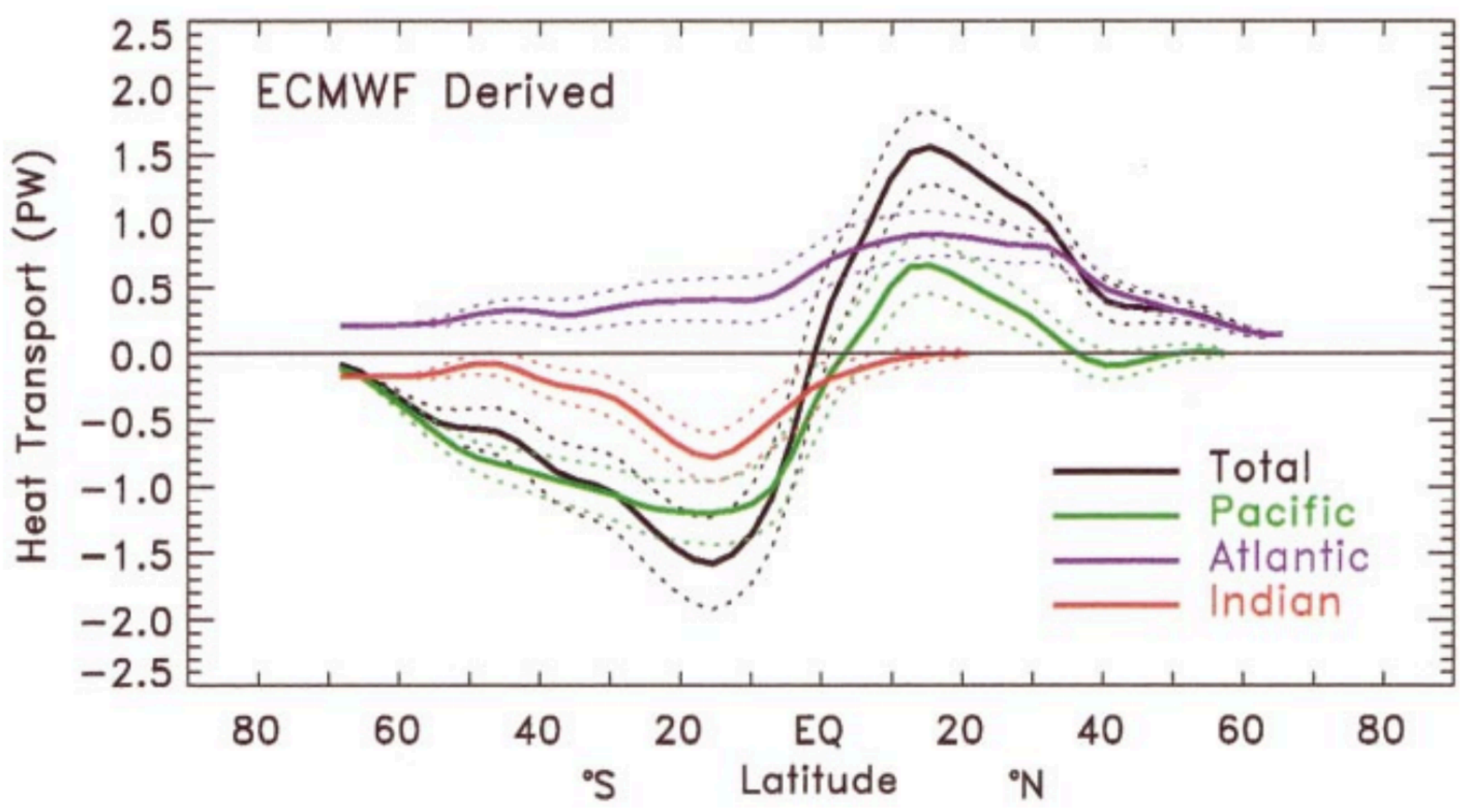
- This happens globally but:
- **Atlantic is unique:** heat transport is northward at all latitudes
- This transport is dominated by the **AMOC** (Atlantic Meridional Overturning Circulation):
  - *Johns et al. (2011)* found **at 26°N nearly linear relationship** between MHT-AMOC:

    0.065 PW/Sv

    Satzbau: Johns found a nearly linear relationship between … at 26 …

  - AMOC carries around ~88% of total MHT in Atlantic

↳ Quelle auch Johns?



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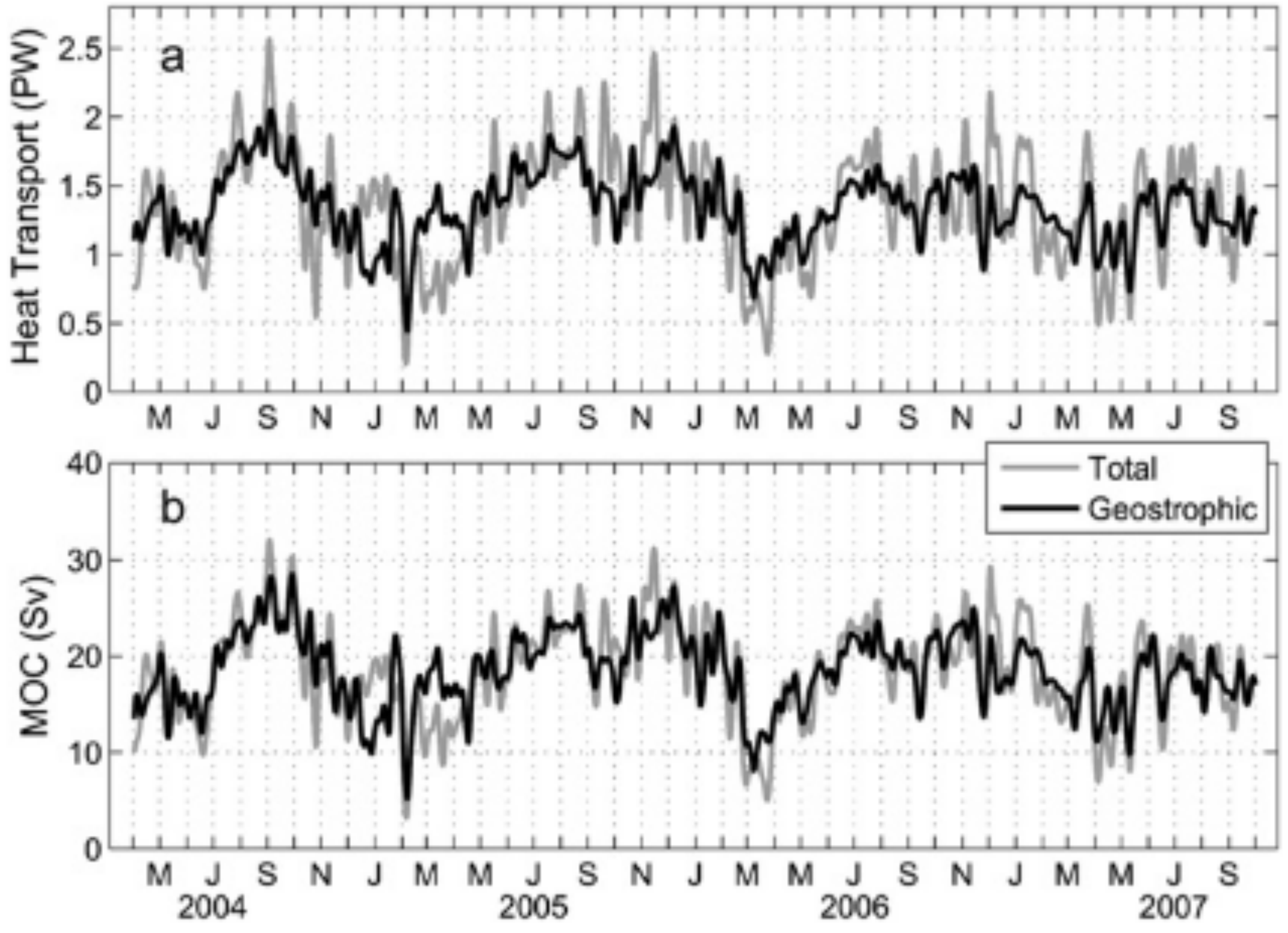


Heat Transport and MOC at 26°N, *Johns et al. (2011)*

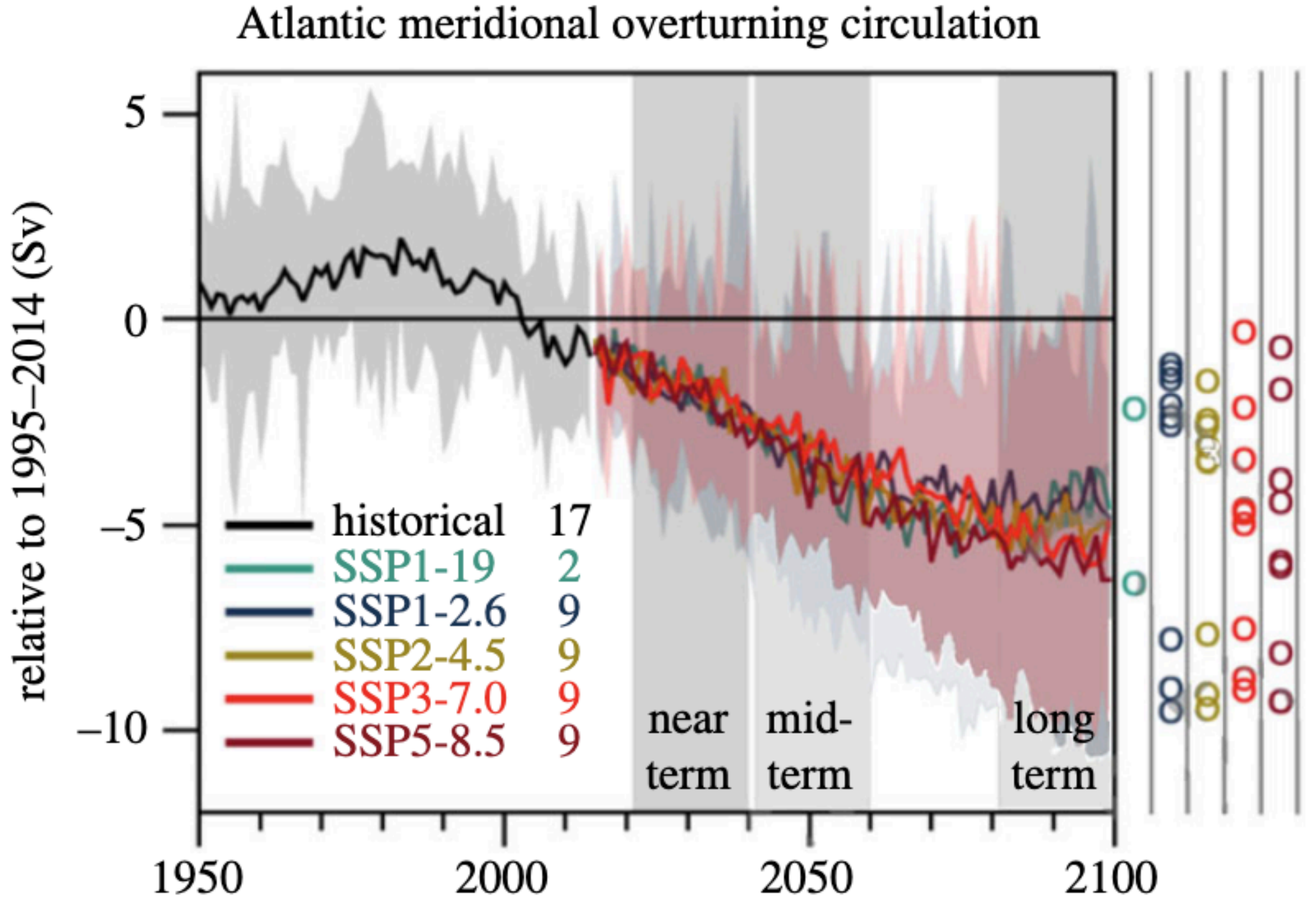
Impacts

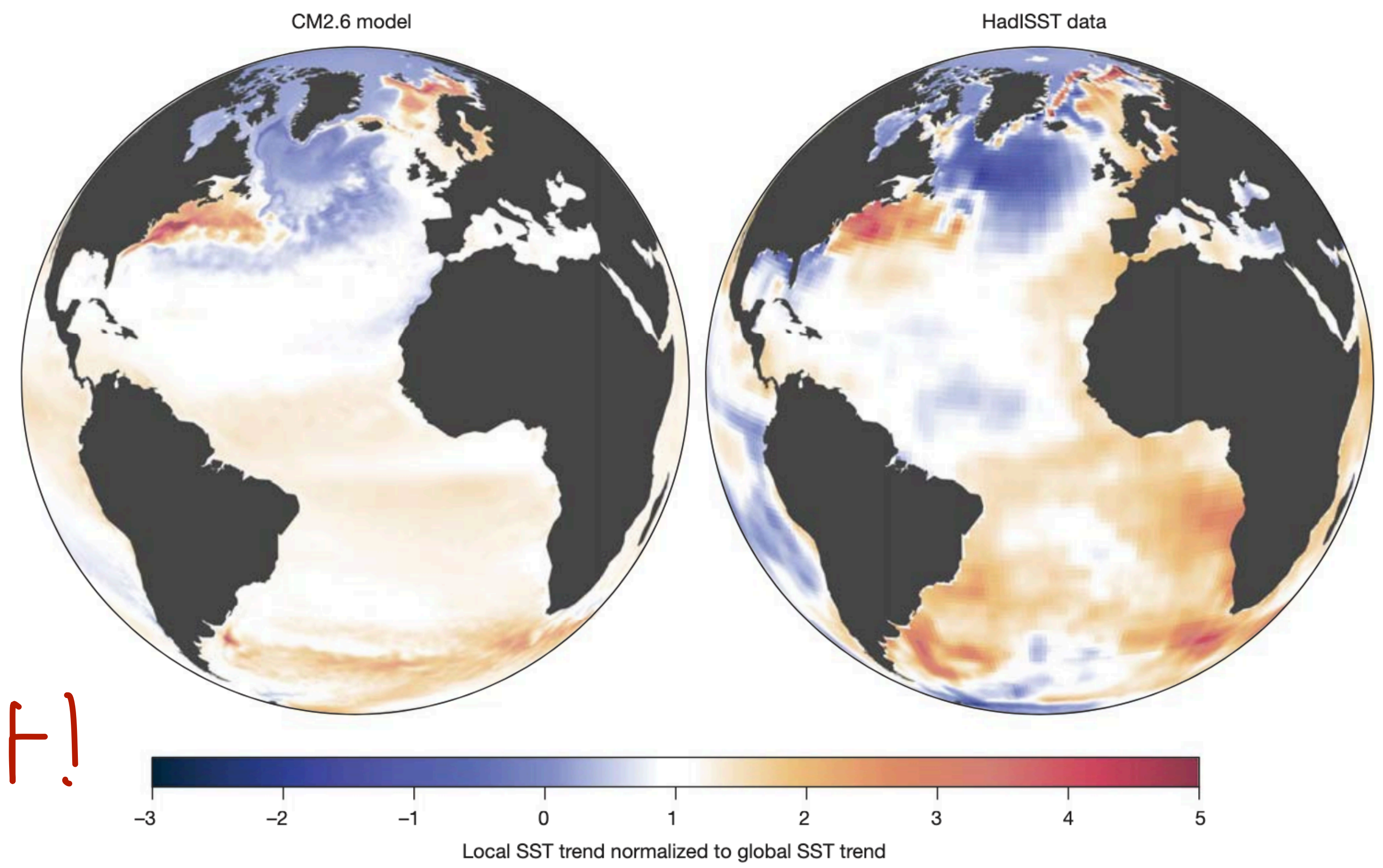
What could happen for weakening AMOC and MHT?

- Climate models predict AMOC weakening in this century Quelle?
- As MHT is directly linked to AMOC → decrease in MHT would be expected
- Weaker MHT can impact: word choice: can impact less heat... ↯ Zb ... could mean
  - Less heat transported northwards and accumulates in the tropics
  - Cooling in the North Atlantic (Cold blob) → colder temperatures over Europe
  - Position of ITCZ ~~can~~ change, shift southward *(Marshall et al. 2013)* ...

Can we confirm this with observational data?

→ Emphasis! Das ist willkommen! Fett, oder Box o.ä.





Figures: upper from *Johns et al. (2023)* (Adapted from IPCC AR6 WG1), lower normalized SST trends from *Caesar et al. (2018)*

Research Gap

What do observations tell us and what remains unknown?

↳ könnte noch etwas pointierter sein?

- Observed 0.17 PW MHT reduction at 26°N for 2009-2016, *Bryden et al (2020)* - but is this a **forced trend or natural variability?**
  ↳ das ist ein Bindestrich (hyphen), du willst ein em-dash (länger)
- Interannual variability driven by **climate modes** - e.g. AMOC/MHT drop in 2009, following record low NAO
- AMOC-MHT connection well-established for 26°N - **basin-wide latitudinal MHT coherence** poorly understood
  auch fett?

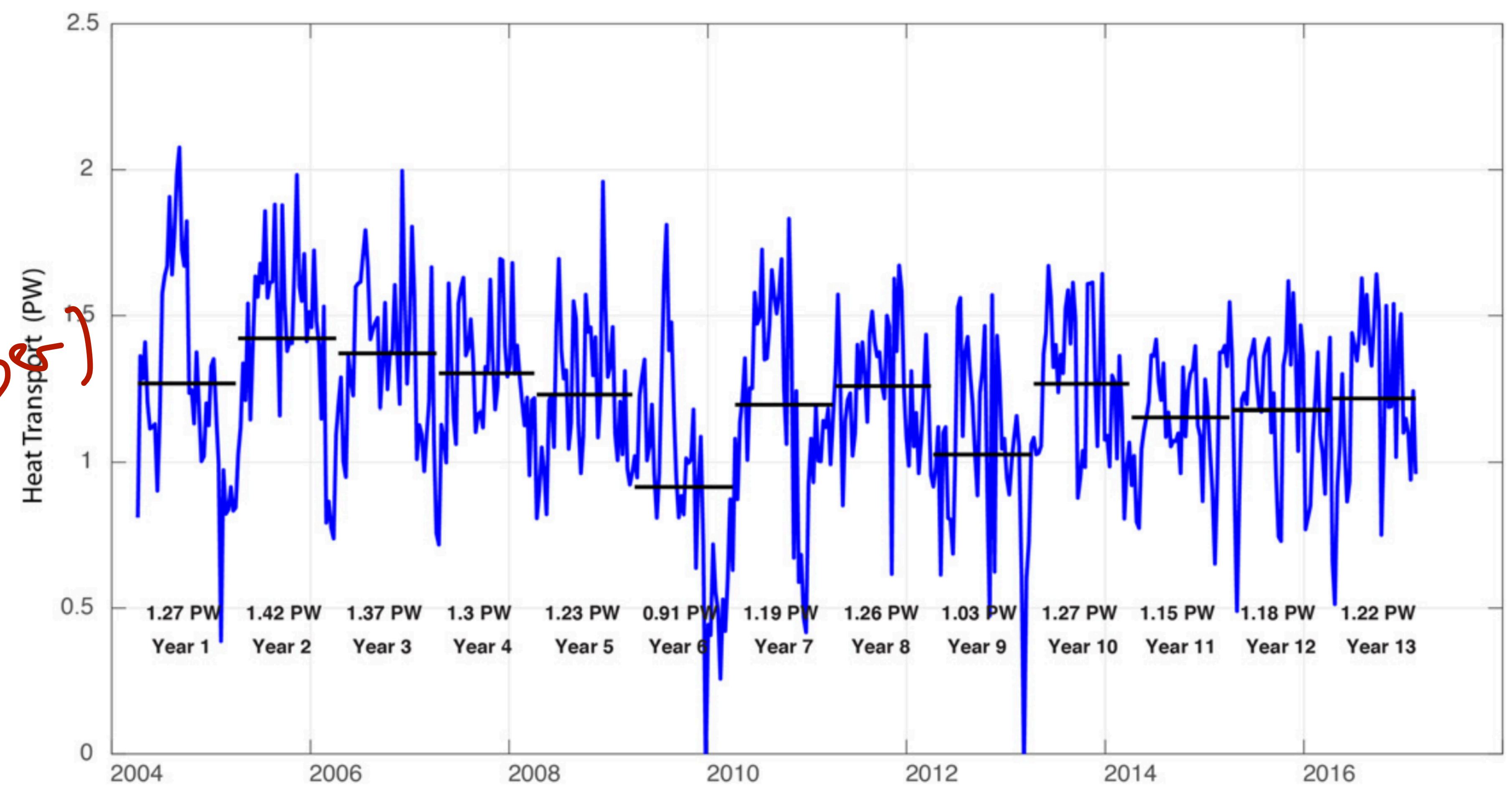


Figure from *Bryden et al (2020)*

Issues: Wortwahl: vielleicht schon direkt Open questions oä?

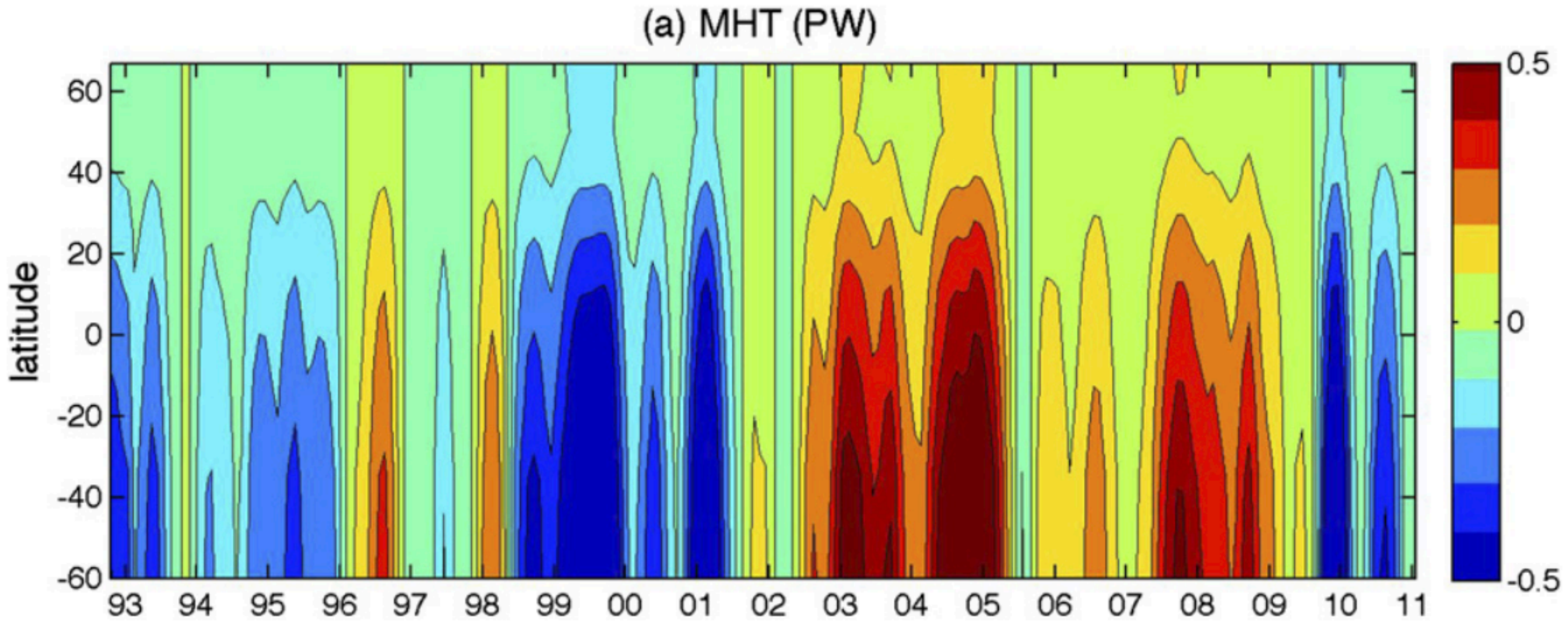
1. **Observational limitations:** continuous mooring arrays since early 2000s, spatially sparse & short time period
   ↳ Formulierung: so klingt es fast positiv m aber du willst ja eigentlich betonen dass das nicht sehr lang ist
2. To be able to **detect anthropogenic forced trend** we first need to understand its **natural variability**

Thesis objectives

Investigating basin-wide coherence of Atlantic MHT

„How" oder „Is MHT coherent..."?

1. How coherent is MHT over the Atlantic?
2. What drives incoherence? Are differences caused by circulation changes or by changes in surface heat flux?



First mode of variability of MHT anomalies (PW), *Kelly et al. (2014)*.

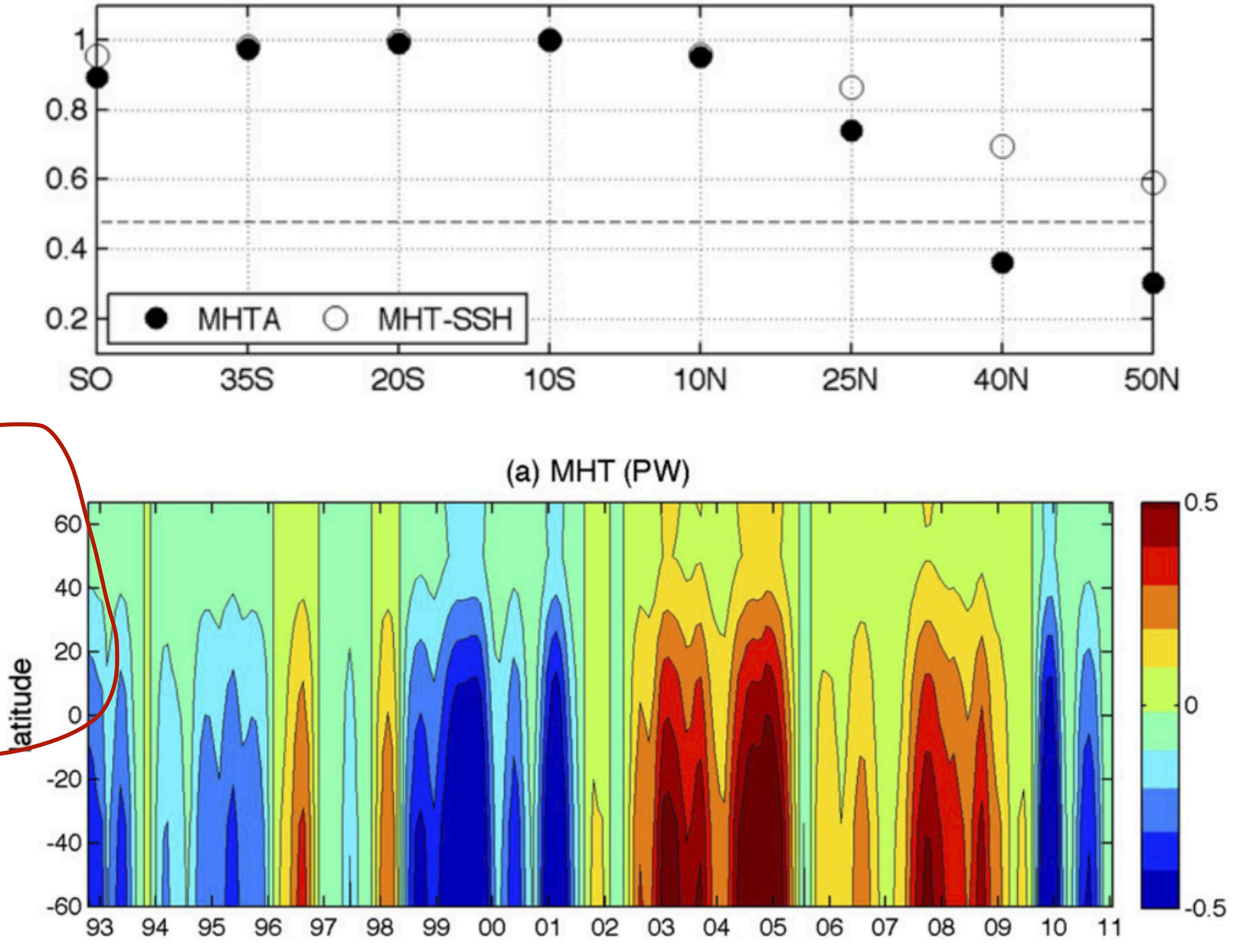
Thesis objectives

Context: what is coherence?

- Coherence: Anomalies covarying across latitude bands
- *Kelly et al. (2014)* found coherence from 35°S until 40°N
  - High correlation between bands
- Anomalies increase southward auch einrücken?
- ~~Dominant timescale is interannual (lower plot)~~

Fett oä!

- Open question: Does the signal truly arise simultaneously or do they propagate through the basin with a lead/lag relationship?
- ~~Their~~ records end 2011 - extend by using *Calafat et al. (2025)* estimates for MHT until 2020, improved uncertainty estimates



Upper: Spatial coherence of MHT, correlation with 10°N, Lower: First mode of variability of MHT anomalies (PW), *Kelly et al. (2014)*.

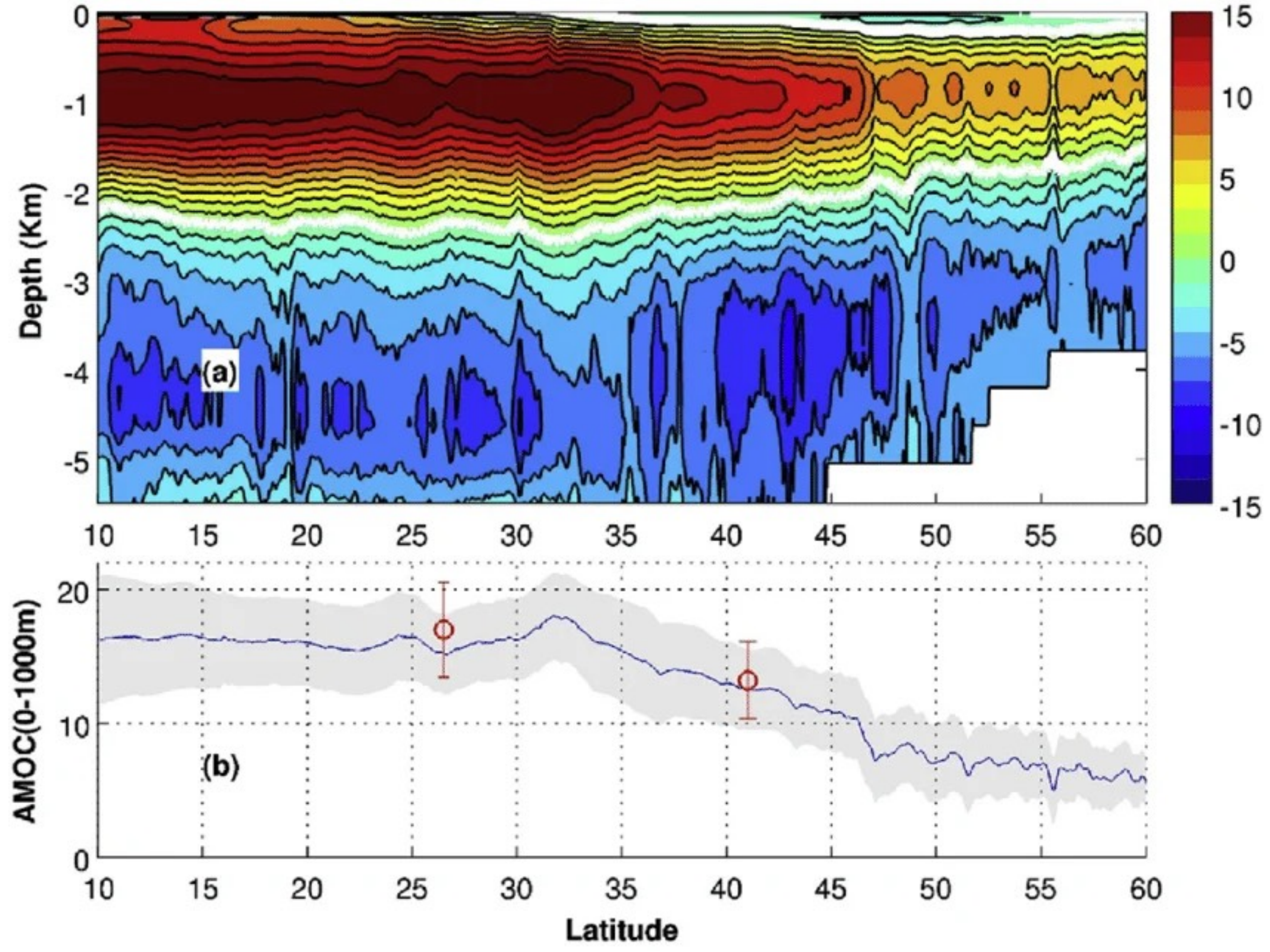
Thesis objectives

Decomposing MHT into its drivers

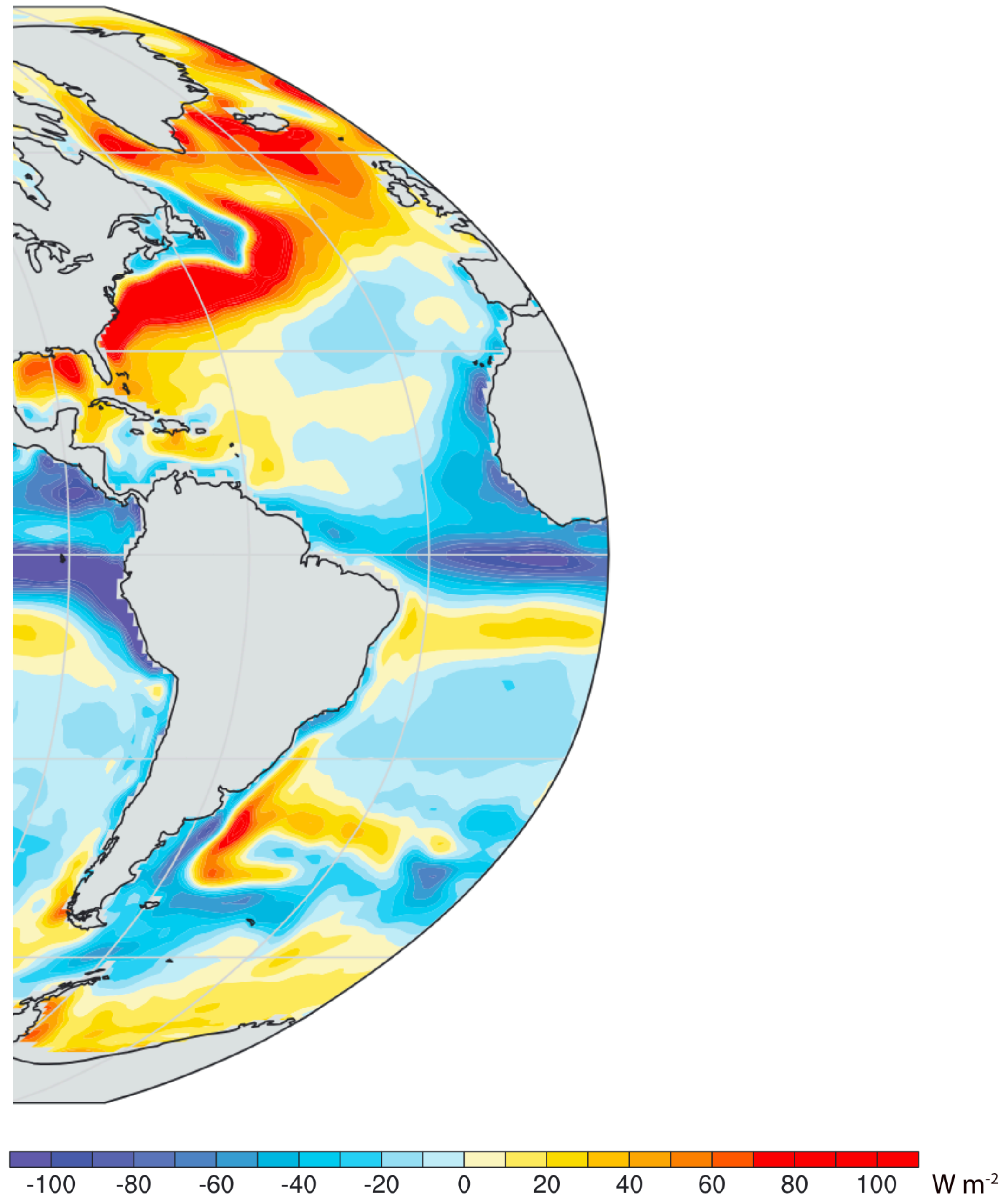
1. How coherent is MHT over the Atlantic?
2. What drives incoherence? Are differences caused by circulation changes or by changes in surface heat flux?

Decomposing MHT into

- Circulation driven and
- Surface heat flux driven components



Mean AMOC Streamfunction and mean transport



Surface heat flux (net upward) for 2000-2014 in W/m^2

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Figure source left: *Wang et al. (2019)*, right: *Trenberth and Fasullo (2017)*

Methods - Data

MHT estimates from the Bayesian Hierarchical Model (BHM)

Warum kursiv?

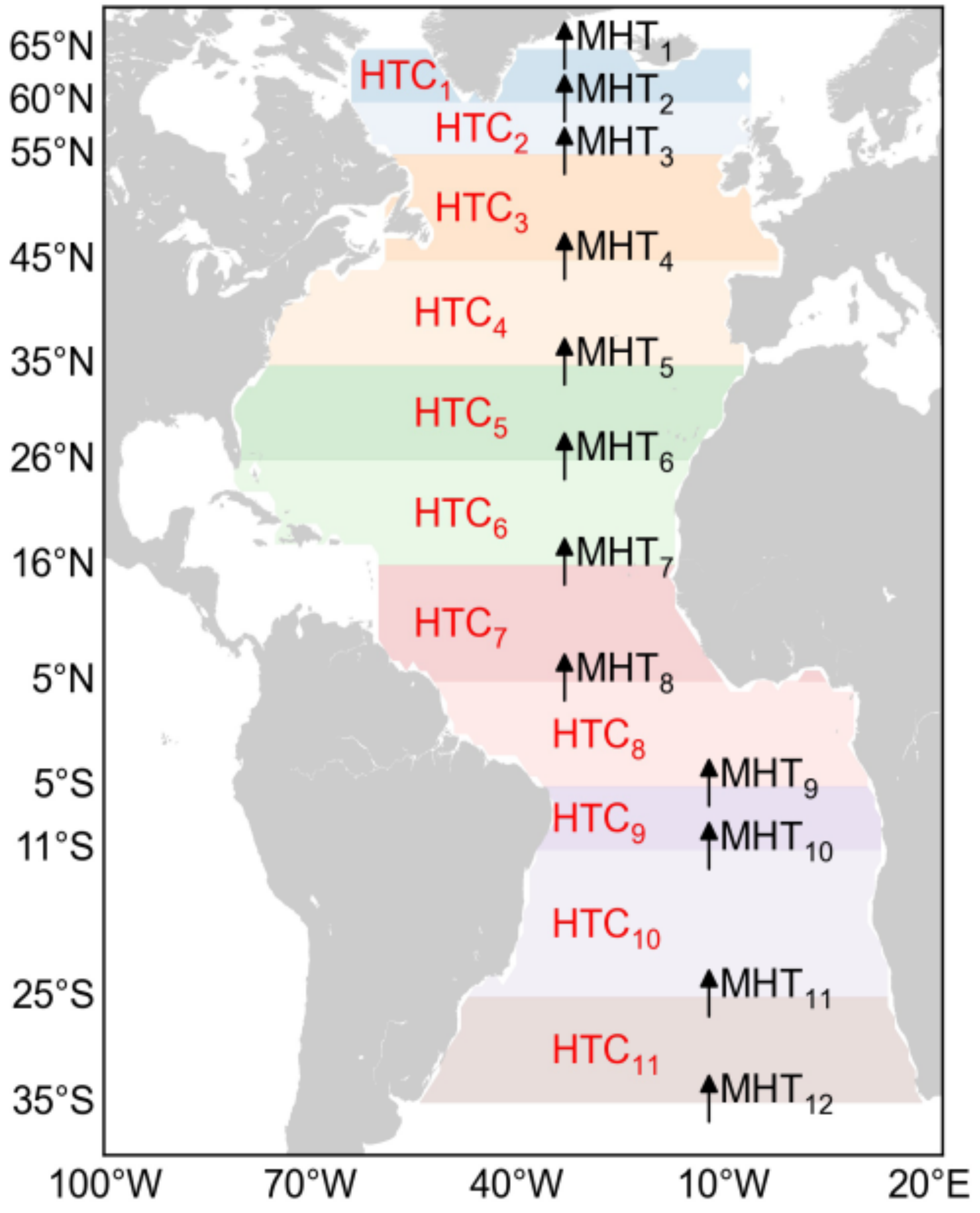
MHT estimates from *Calafat et al. (2025)*:

- Combines **hydrographic data** (argo) with **satellite data** (altimeter and gravimetry)

  Wortwahl

  → Cheaper than mooring arrays, higher latitudinal resolution
- Different datasets are combined by using **Bayesian Hierarchical Model**

  → Accounts for different spatial resolutions between datasets and **uncertainty propagation**
- Dataset offers:
  - Probabilistic estimates for MHT and Heat Transport Convergence (HTC) for
  - **12 latitude bands** from **35°S to 65°N**
  - From 2004-2020 at 3-month resolution



12 Latitude bands, *Calafat et al. (2025)*

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Methods - Data

How is MHT calculated using the BHM?

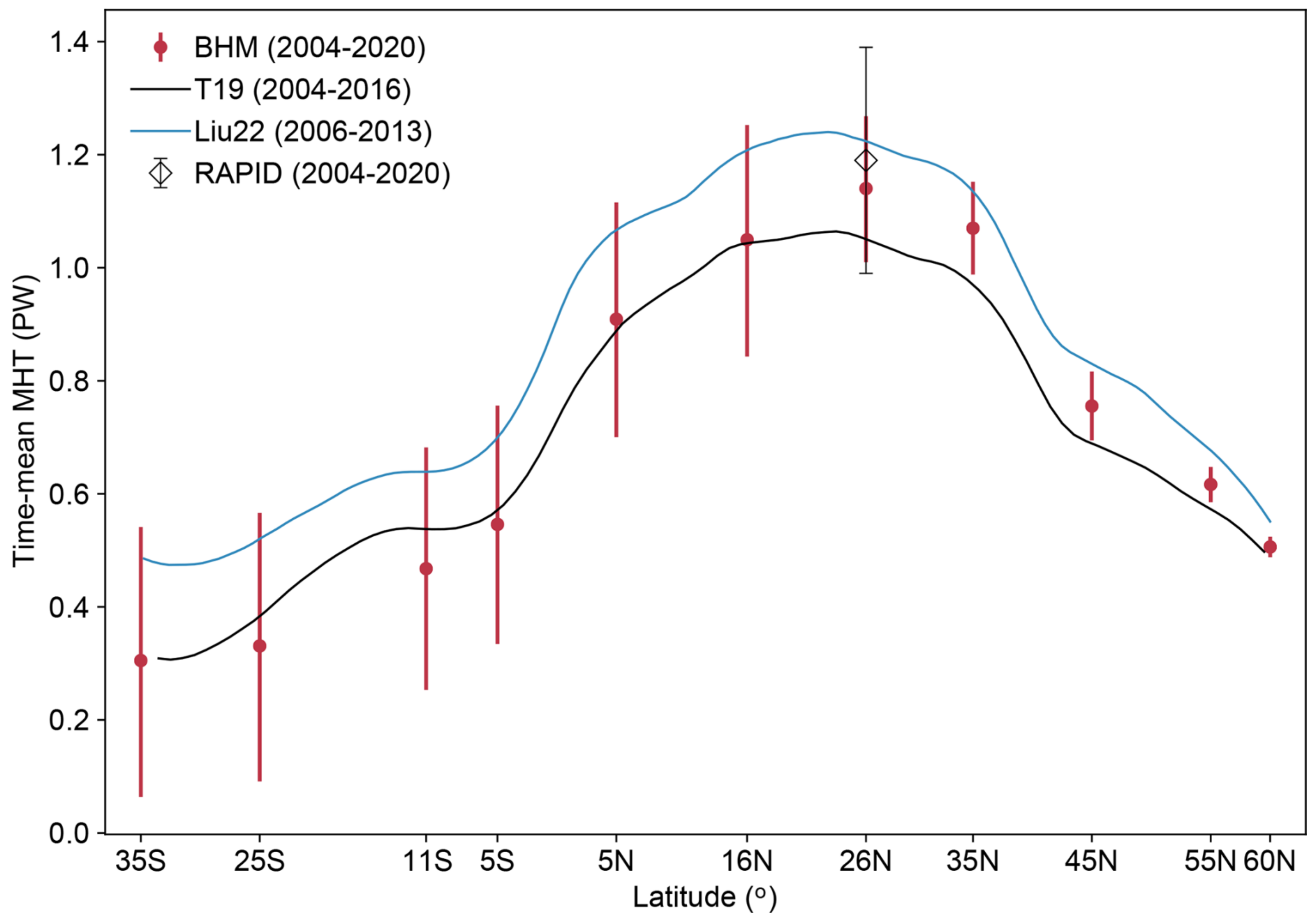
- MHT indirectly inferred by meridionally integrating HTC

$$\mathrm{MHT}_{i,t} = \mathrm{MHT}_{1,t} + \sum_{j=1}^{i-1} \mathrm{HTC}_{j,t},\ i = 1, \ldots, M+1,$$

- HTC is computed as the residual of
  Ocean Heat Content (OHC) tendency - surface heat flux (F_S)

Comparing to other datasets:

- BHM estimates capture MHT well (e.g. at 26°N)
- comparably to other methods



Calafat et al. (2025)

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→ Wein

First look at the Data

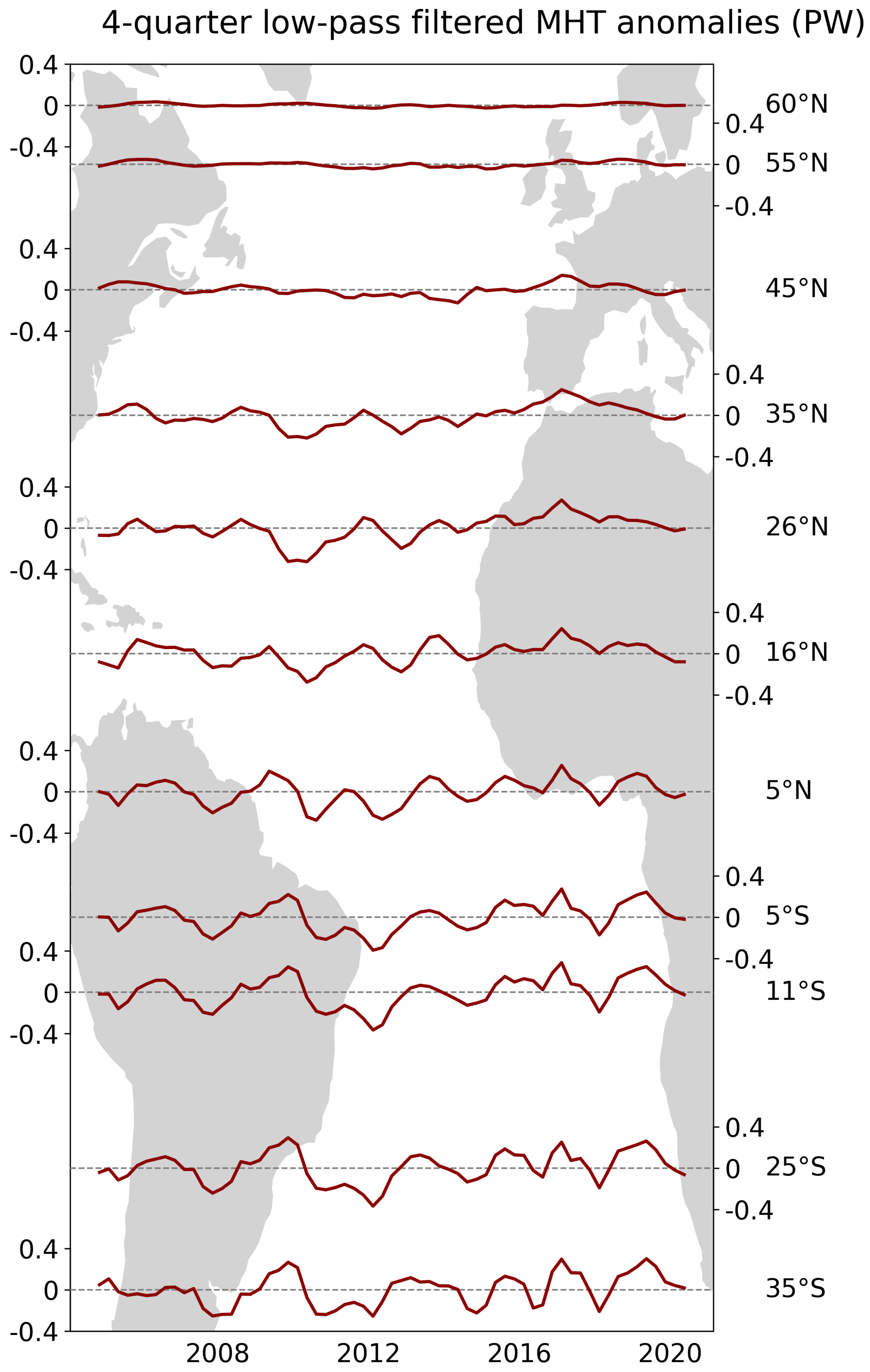
Processing and initial observations

Processing the data:

- Deseasonalized
- 4-quarter low-pass filtered (running mean)

First glance impressions:

- Weakest signal at northern most latitudes
- The more south the higher the variability and the uncertainty



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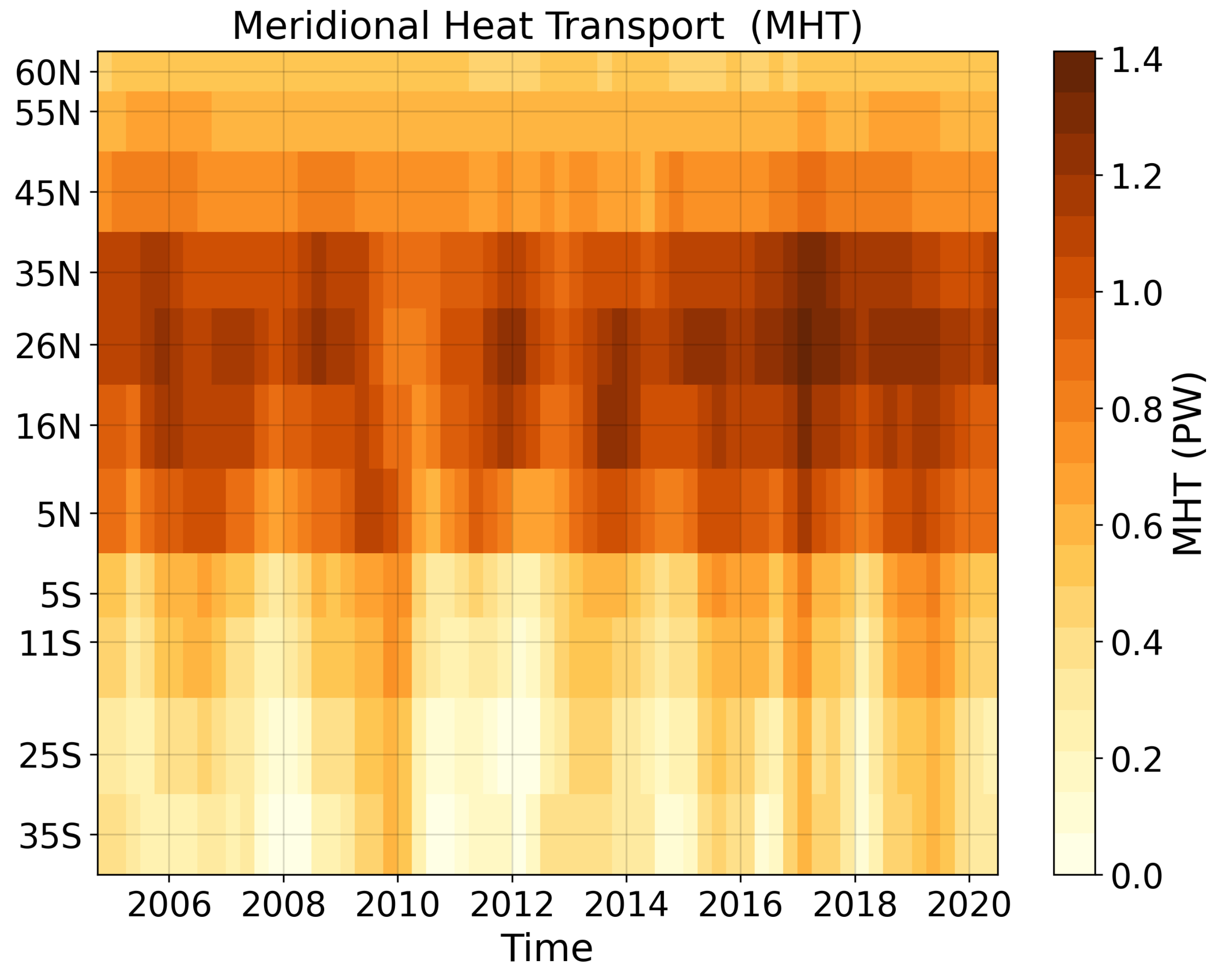
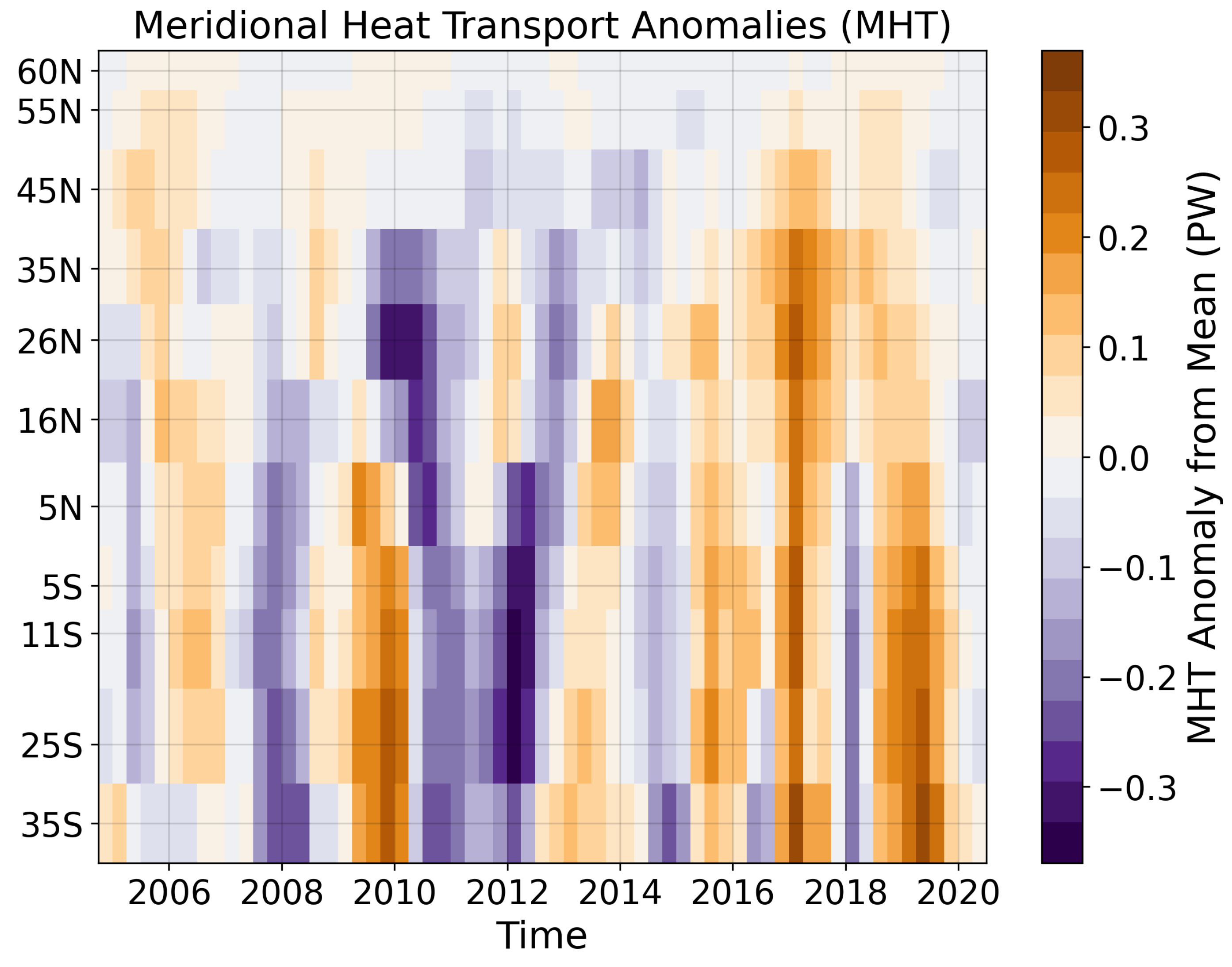
Reproducing Calafat et al. (2025)

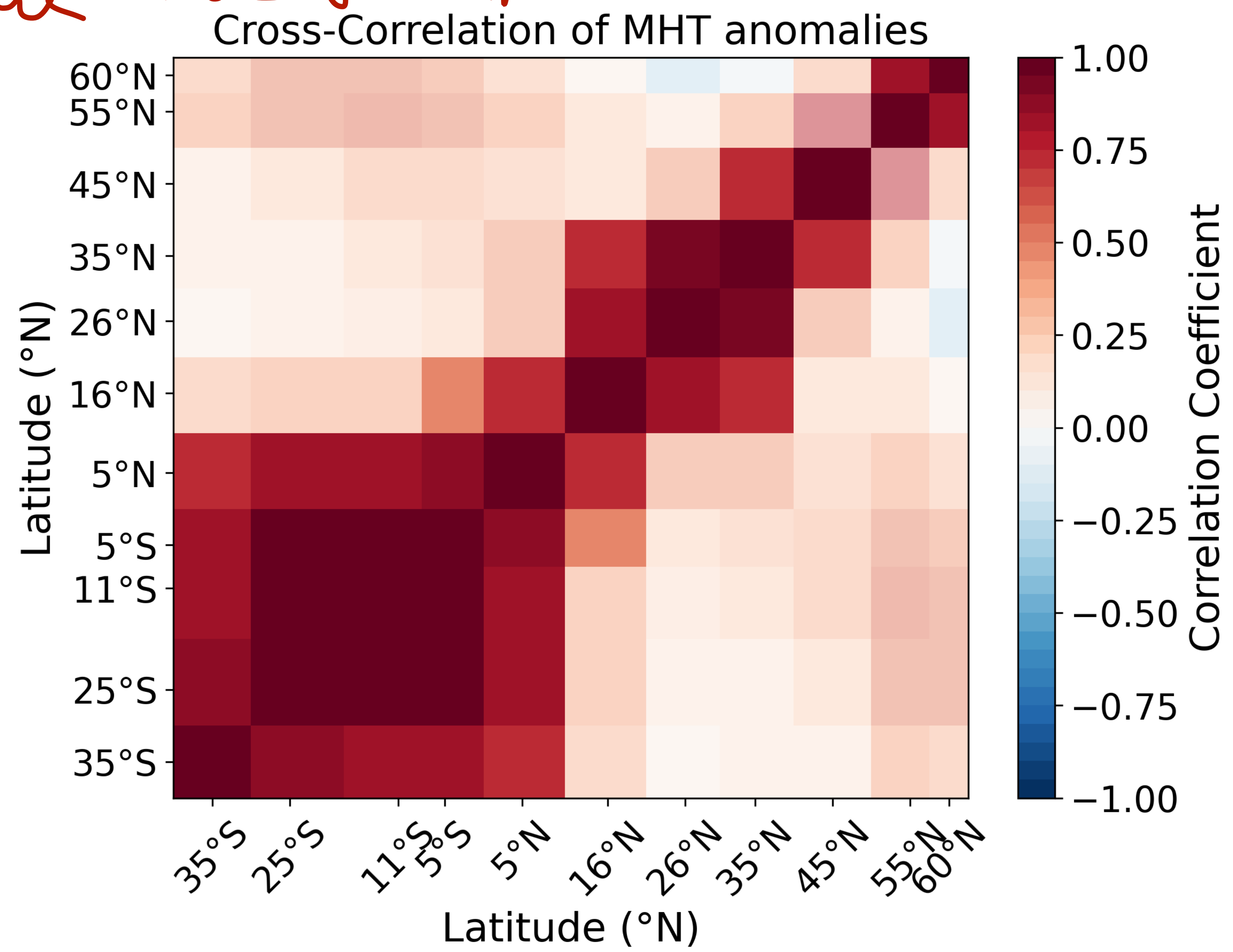
Two covarying latitude bands: 35 - 16°N and 5°N - 35°S

- Anomalies are computed for each latitude band's individual time mean
- Strongest MHT over tropics, maximum at 26°N
- Smallest anomalies at northern latitudes - likely reflecting the MHT=0 constraint at 65°N
- Basin wide pattern (coherence) for specific years

→ sind das schon zu viele Ergebnisse?
siehe Nachricht



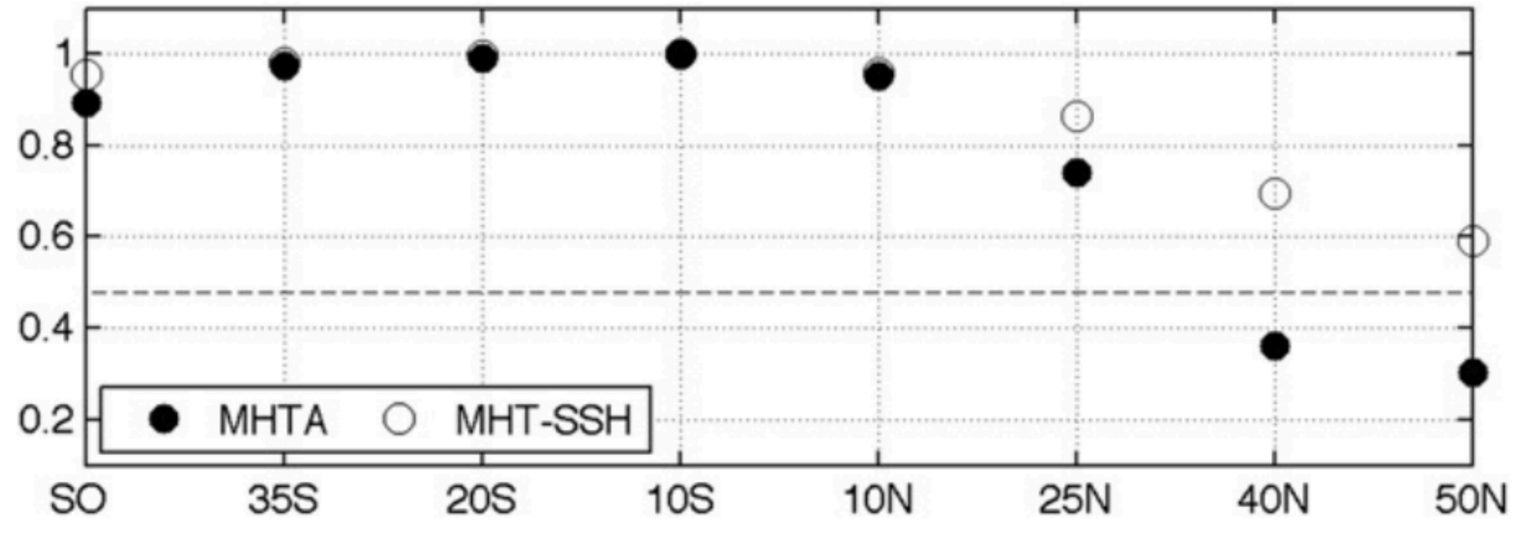




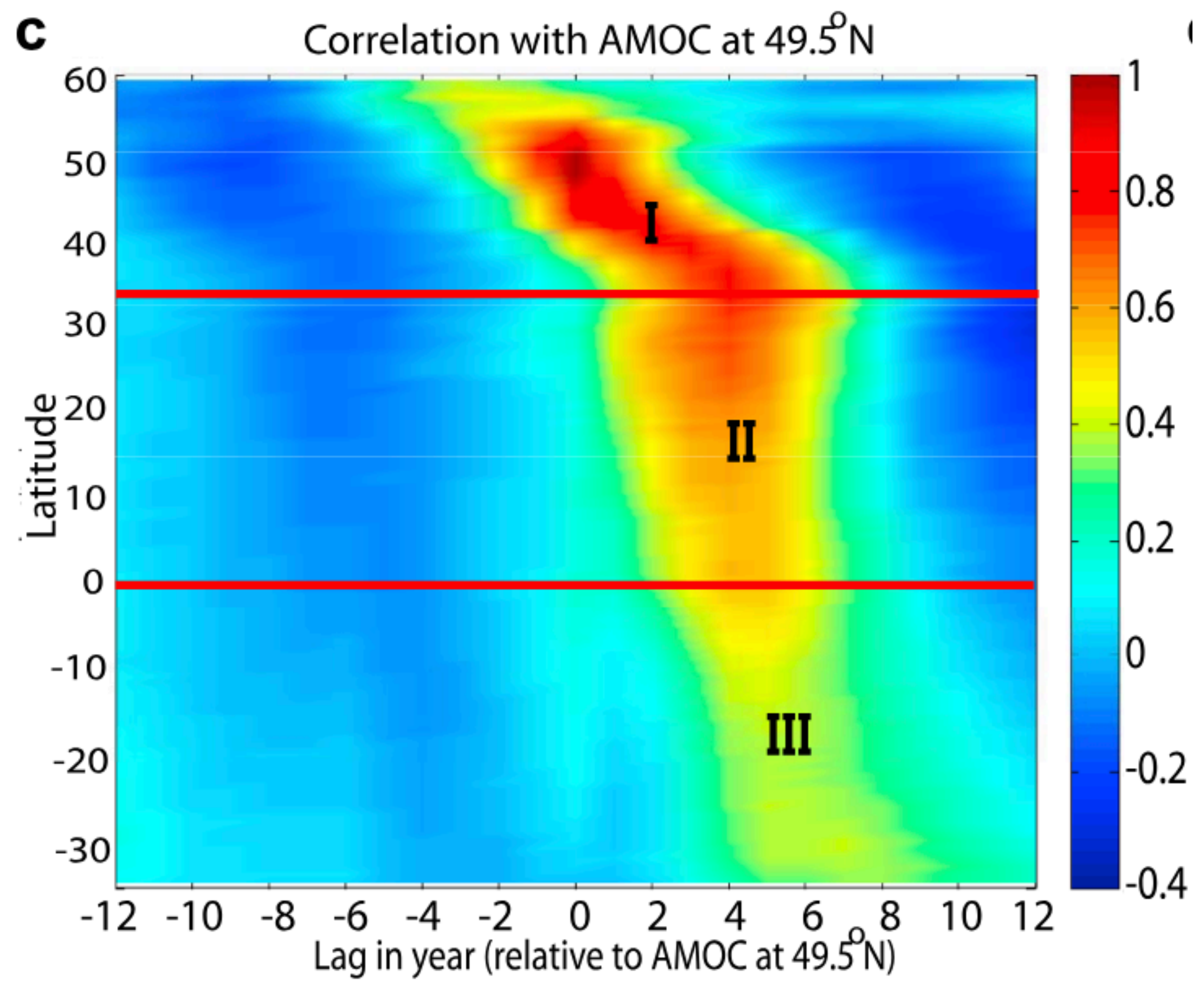
Analysis approach and next steps

Investigating basin-wide coherence of Atlantic MHT

- 0-lag analysis (*Kelly et al. (2014)*) for coherence
- Lead/lag analysis: signal propagation between latitude bands
  - lead/lag correlation for each reference latitude (*Zhang et al. (2010)* approach)
- Including significance testing
- Decompose MHT variability into circulation and surface heat flux contributions
- Correlate to climate indices (e.g. NAO)



Spatial coherence of MHT, correlation with 10°N, *Kelly et al. (2014)*.



Correlation between AMOC variations at 49.5°N and AMOC variations at all latitudes over the entire control simulation, *Zhang et al. (2010)*.

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